

SOC BASELINE FRAMEWORK MATRIX					GENERAL
Domain	Baseline Components	Objective	Example	Component Function	
INFRASTRUCTURE & TOOLS	• SIEM platform • EDR/XDR • Threat Intelligence feeds • Network sensors (NDR/IDS/IPS) • SOAR automation • Case/Ticketing system	To establish monitoring, detection and response capabilities through integrated tools.	Splunk, QRadar, Cortex XDR, BrightCloud TI, Cortex XSOAR	SIEM: Collects and correlates logs for alerts. EDR/XDR: Detects and contains endpoint threats. TI feeds: Adds context with IOC enrichment. NDR/IDS/IPS: Detects malicious network patterns. SOAR: Automates alert triage and response. Ticketing: Tracks incidents and documentation.	
LOG SOURCE COVERAGE	• Firewall • Proxy • Endpoint • AD / DNS • Cloud / VPN / WAF • Email Gateway	Ensure comprehensive visibility across all network and system layers.	Coverage mapped to MITRE ATT&CK data sources	Firewall: Detect inbound/outbound threats. Proxy: Logs user web activity and filtering. Endpoint: Captures host behaviour and infection traces. AD/DNS: Identifies authentication abuse and resolution anomalies. Cloud/VPN/WAF: Monitors access and API security. Email Gateway: Detects phishing and malicious attachments.	
OPERATIONAL PROCESS	• Incident Response Process (NIST/SANS) • Runbooks & Playbooks • Shift Handover Procedure • Escalation Matrix • Reporting Templates	Define standard operating procedures for consistent and traceable SOC operations.	NIST SP 800-61 Rev.2, SANS PICERL	IR Process: Guides step-by-step handling of incidents. Runbooks/Playbooks: Ensures consistent technical response. Shift Handover: Maintains investigation continuity. Escalation Matrix: Defines authority and escalation chain. Reports: Summarises incident trends and performance.	
PEOPLE & CAPABILITY	• Tier 1: Monitoring & Triage • Tier 2: Investigation & Response • Tier 3: Threat Hunting & IR • SOC Manager / Shift Lead	Establish clear roles, responsibilities and skill levels for efficient incident management.	MITRE ATT&CK awareness, OSINT, SIEM mastery	Tier 1: Monitors alerts, validates true/false positives. Tier 2: Performs deep investigation and containment. Tier 3: Conducts forensics, hunting and IR. SOC Manager: Oversees operations, ensures SLA compliance.	
GOVERNANCE & COMPLIANCE	• SOC Policy & SOP • Access Control & Segregation • Severity Classification • KPI Metrics & SLAs • Audit & Review	Align SOC operations with regulatory and industry standards.	RMIT 10.12, ISO 27001 A.16, NIST CSF	Policy/SOP: Defines SOC rules and responsibilities. Access Control: Limits privileges and enforces SoD. Severity Classification: Guides prioritisation and reporting. KPI/SLA: Measures performance and responsiveness. Audit/Review: Ensures compliance and continual improvement.	
DETECTION & RESPONSE	• Use Case Library • Alert Prioritisation Model • Correlation Rules • Incident Severity Criteria	Ensure rapid, accurate and prioritised detection aligned with business risk.	MITRE ATT&CK technique mapping	Use Cases: Define what threats to detect. Prioritisation Model: Determines alert urgency. Correlation Rules: Combine multi-log patterns into a single event. Severity Criteria: Assigns risk impact to incidents.	
THREAT INTELLIGENCE INTEGRATION	• Internal & External TI Feeds • IOC Enrichment Workflow • Threat Actor Profiling	Enhance situational awareness and proactive threat identification.	BrightCloud, MISP, Anomali, Group-IB	TI Feeds: Supply up-to-date indicators of compromise. Enrichment: Adds threat context to alerts automatically. Actor Profiling: Links activity to known groups and campaigns.	
REPORTING & COMMUNICATION	• Daily/Weekly/Monthly SOC Reports • Client Notification SOP • Management Dashboard • SLA Metrics	Maintain transparency, accountability and situational awareness across stakeholders.	SOC dashboard (Power BI / Splunk), incident summaries	Reports: Track alert volumes and trends. Client SOP: Ensures timely stakeholder updates. Dashboard: Provides real-time visibility of SOC health. SLA Metrics: Demonstrates SOC performance and efficiency.	
ENVIRONMENT SECURITY & MAINTENANCE	• SOC System Hardening • Backup & DR • NTP Time Sync • Asset Inventory • Environment Segregation	Protect SOC systems from internal and external compromise and ensure resilience.	RMIT 10.12 – Environment Segregation	System Hardening: Reduces SOC system vulnerabilities. Backup/DR: Preserves data and ensures continuity. Time Sync: Maintains log integrity and accuracy. Inventory: Tracks monitored assets and coverage. Segregation: Prevents cross-environment contamination.	
CONTINUOUS IMPROVEMENT	• Threat Hunting • Use Case Review • Lessons Learned • Red/Blue Team Exercises • Training & Certification	Drive ongoing capability development and maturity growth.	Monthly RCA reviews, CTF, purple teaming	Threat Hunting: Identifies undetected threats. Use Case Review: Refines detection coverage. Lessons Learned: Feeds improvement from past incidents. Red/Blue Team: Tests detection and response readiness. Training: Builds analyst skill maturity.	